



UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO

FACULTAD DE INGENIERÍA

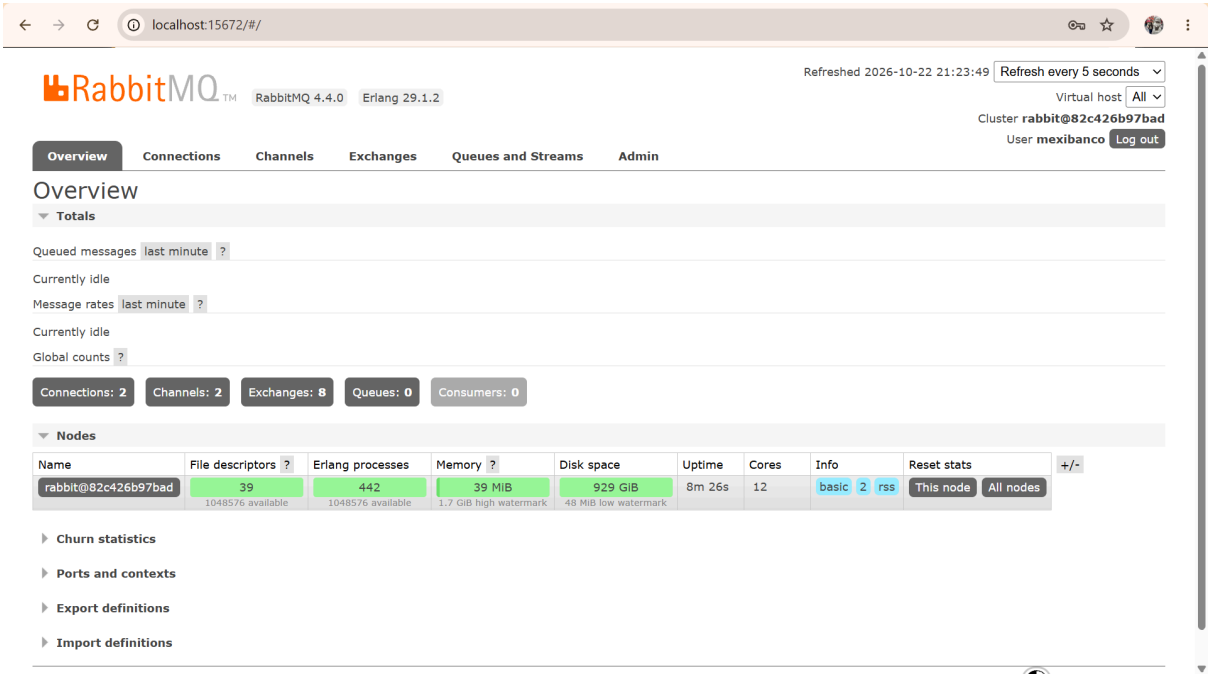
SISTEMAS DISTRIBUIDOS

LABORATORIO SESIÓN 9

CONTRERAS DEL ÁNGEL DIEGO ADRIÁN

SEMESTRE 2027-1

24 DE OCTUBRE DE 2026



Panel de administración de RabbitMQ antes de empezar. Aparecen 0 colas y 0 consumidores no se ha declarado nada todavía, así que no hay dónde guardar mensajes.

```
PS C:\Users\dieg7\mexi-banco-s09> Invoke-RestMethod -Method Post -Uri http://localhost:18080/api/cuentas -ContentType "a
pplication/json" -Body '{"clabe":"002180000000000001","titular":"Ana","saldoInicial":1000}'

id clabe              titular saldo
-- -----              ------- -----
 3 002180000000000001 Ana      1000


PS C:\Users\dieg7\mexi-banco-s09> Invoke-RestMethod -Method Post -Uri http://localhost:18080/api/cuentas -ContentType "a
pplication/json" -Body '{"clabe":"002180000000000002","titular":"Beto","saldoInicial":500}'

id clabe              titular saldo
-- -----              ------- -----
 4 002180000000000002 Beto      500
```

Se crean las cuentas que se van a usar de prueba Ana con saldo 1000 y Beto con saldo 500, para ello utilicé el puerto 18080.

```
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s http://localhost:18080/api/transferencias
{"type":"about:blank","title":"Bad Request","status":400,"detail":"Required parameter 'estado' is not present.","instanc
e":"/transferencias"}
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s http://localhost:18080/api/cuentas/002180000000000001
{"id":3,"clabe":"002180000000000001","titular":"Ana","saldo":950.00}
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s http://localhost:18080/api/cuentas/002180000000000002
{"id":4,"clabe":"002180000000000002","titular":"Beto","saldo":550.00}
PS C:\Users\dieg7\mexi-banco-s09>
```

Saldos finales tras transferir 50 de Ana a Beto. Ana se quedó con 950 y Beto con 550, así que el dinero sí se movió.

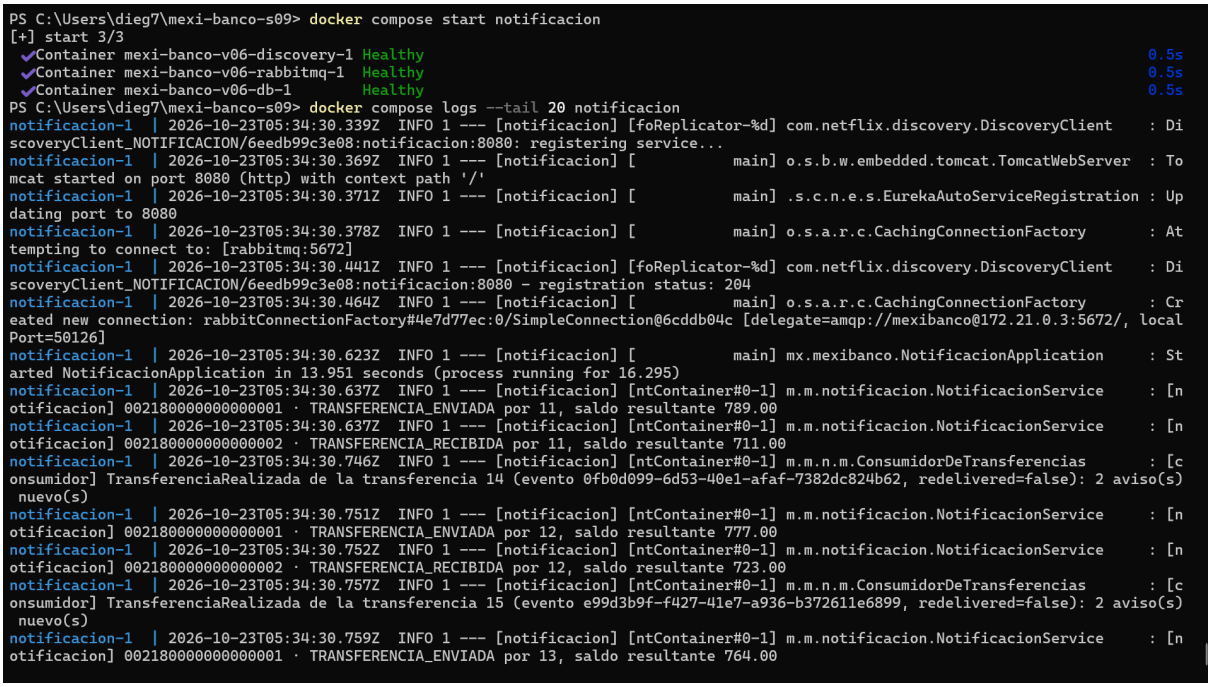
```
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -i -X POST http://localhost:18080/api/transferencias -H "Content-Type: applic
ation/json" -d "@t50.json"
HTTP/1.1 201 Created
transfer-encoding: chunked
Content-Type: application/json
Date: Fri, 23 Oct 2026 05:07:49 GMT

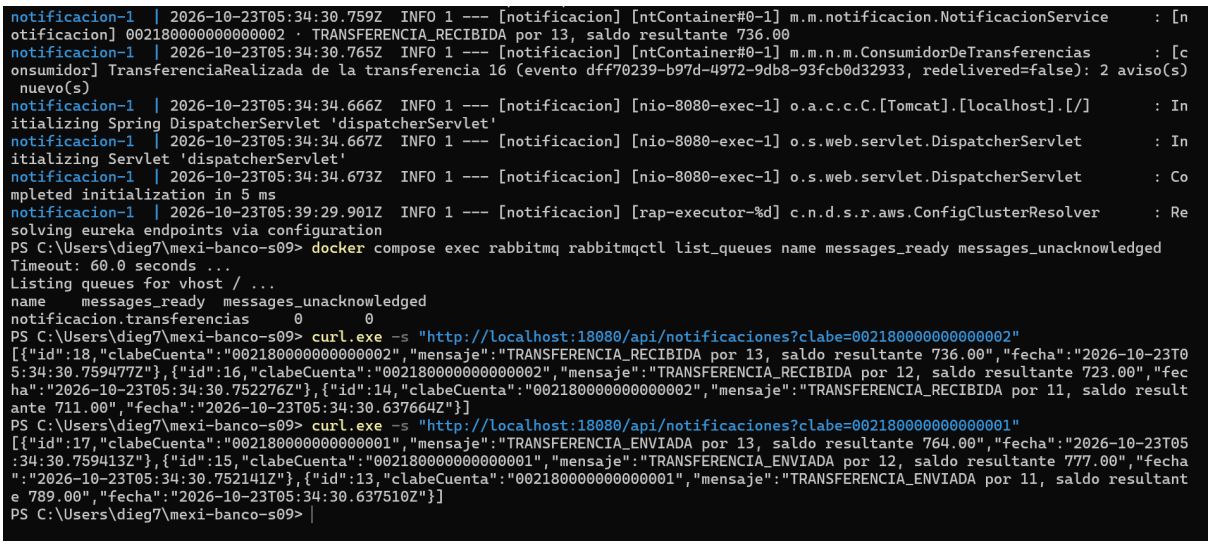
{"id":13,"claveOrigen":"002180000000000001","eventoId":"e7f7d246-293f-45f0-9bce-d02f7c539620","claveDestino":"0021800000
00000002","monto":50,"fecha":"2026-10-23T05:07:48.584904045Z","estado":"COMPLETADA","motivo":null,"saldoOrigenTrasCargo"
:800.00,"intentosDeCompensacion":0,"actualizada":"2026-10-23T05:07:49.651946257Z"}
PS C:\Users\dieg7\mexi-banco-s09> docker compose logs --tail 20 transferencia | Select-String "evento"

transferencia-1  | 2026-10-23T05:07:49.763Z ERROR 1 --- [transferencia] [nio-8080-exec-8]
m.m.transferencia.TransferenciaService   : Transferencia 13: quedo terminada pero el evento TransferenciaRealizada
e7f7d246-293f-45f0-9bce-d02f7c539620 NO se publico: Ninguna cola recibio el evento
e7f7d246-293f-45f0-9bce-d02f7c539620 con la routing key transferencia.realizada (NO_ROUTE)
```

La transferencia responde 201 y queda completada, y la respuesta ya trae su eventoId. En el log, el broker avisa que ninguna cola recibió el evento porque todavía no existe una cola enlazada. La transferencia no falla, solo se pierde ese aviso, y la mandatory es la que lo hace visible.

```
PS C:\Users\dieg7\mexi-banco-s09> docker compose stop notificacion
[+] stop 1/1
 ✔Container mexi-banco-v06-notificacion-1 Stopped                                                          3.8s
PS C:\Users\dieg7\mexi-banco-s09> foreach ($m in 11,12,13) {
>>   @{claveOrigen="002180000000000001"; claveDestino="002180000000000002"; monto=$m} | ConvertTo-Json -Compress | Set-Content -Encod
ing ascii "t$m.json"
>>   curl.exe -s -o NUL -w '%{http_code}\n' -X POST http://localhost:18080/api/transferencias -H "Content-Type: application/json" -d
"@t$m.json"
>> }
201
201
201
PS C:\Users\dieg7\mexi-banco-s09> docker compose exec rabbitmq rabbitmqctl list_queues name messages_ready messages_unacknowledged
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages_ready  messages_unacknowledged
notificacion.transferencias     3       0
PS C:\Users\dieg7\mexi-banco-s09> docker compose start notificacion
[+] start 4/4
 ✔Container mexi-banco-v06-db-1           Healthy                                                          0.5s
 ✔Container mexi-banco-v06-discovery-1    Healthy                                                          0.5s
 ✔Container mexi-banco-v06-rabbitmq-1     Healthy                                                          0.5s
 ✔Container mexi-banco-v06-notificacion-1 Started                                                          0.7s
PS C:\Users\dieg7\mexi-banco-s09> docker compose logs --tail 20 notificacion
notificacion-1  |   '  |____| .__|_| |_|_| |_\__, | / / / /
notificacion-1  |  =========|_|==============|___/=/_/_/_/
notificacion-1  |
notificacion-1  |  :: Spring Boot ::                (v3.3.4)
notificacion-1  |
notificacion-1  | 2026-10-23T05:34:17.586Z  INFO 1 --- [notificacion] [           main] mx.mexibanco.NotificacionApplication     : St
arting NotificacionApplication v0.9.0 using Java 21.0.12.1 with PID 1 (/app/app.jar started by root in /app)
notificacion-1  | 2026-10-23T05:34:17.591Z  INFO 1 --- [notificacion] [           main] mx.mexibanco.NotificacionApplication     : No
 active profile set, falling back to 1 default profile: "default"
notificacion-1  | 2026-10-23T05:34:20.391Z  INFO 1 --- [notificacion] [           main] .s.d.r.c.RepositoryConfigurationDelegate : Bo
```





Con notificación apagada se hacen tres transferencias (11, 12 y 13) y las tres responden 201. La cola notificacion.transferencias guarda 3 mensajes listos y 0 sin confirmar el recado espera aunque el consumidor no esté vivo. Al volver a encender, notificación lee los tres eventos acumulados, cada uno genera 2 avisos, uno para quien envió y otro para quien recibió, con redelivered=false, es decir, primera entrega. La cola bajó a 0 mensajes. La bandeja de Beto muestra TRANSFERENCIA_RECIBIDA por 11, 12 y 13, y la de Ana los tres TRANSFERENCIA_ENVIADA.

```
PS C:\Users\dieg7\mexi-banco-s09> $env:FALLAR_ABONO_A_CLABE = "002180000000000002"
PS C:\Users\dieg7\mexi-banco-s09> docker compose up -d transferencia
[+] up 4/4
 ✔Container mexi-banco-v06-discovery-1     Healthy                                                                    7.7s
 ✔Container mexi-banco-v06-db-1            Healthy                                                                    7.7s
 ✔Container mexi-banco-v06-rabbitmq-1      Healthy                                                                    7.7s
 ✔Container mexi-banco-v06-transferencia-1 Started                                                                   10.1s
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s http://localhost:18080/api/cuentas/002180000000000001
{"id":3,"clabe":"002180000000000001","titular":"Ana","saldo":764.00}
PS C:\Users\dieg7\mexi-banco-s09> '{"claveOrigen":"002180000000000001","claveDestino":"002180000000000002","monto":150}' | Set-Conten
t -Encoding ascii t150.json
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -i -X POST http://localhost:18080/api/transferencias -H "Content-Type: application/json" -
d "@t150.json"
HTTP/1.1 503 Service Unavailable
Content-Type: application/json
Content-Length: 145

{"timestamp":"2026-10-23T15:37:47.399+00:00","path":"/api/transferencias","status":503,"error":"Service Unavailable","requestId":"bd5
98737-1902"}
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s http://localhost:18080/api/cuentas/002180000000000001
{"id":3,"clabe":"002180000000000001","titular":"Ana","saldo":764.00}
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s "http://localhost:18080/api/notificaciones?clabe=002180000000000001"
[{"id":17,"clabeCuenta":"002180000000000001","mensaje":"TRANSFERENCIA_ENVIADA por 13, saldo resultante 764.00","fecha":"2026-10-23T05
:34:30.759413Z"},{"id":15,"clabeCuenta":"002180000000000001","mensaje":"TRANSFERENCIA_ENVIADA por 12, saldo resultante 777.00","fecha
":"2026-10-23T05:34:30.752141Z"},{"id":13,"clabeCuenta":"002180000000000001","mensaje":"TRANSFERENCIA_ENVIADA por 11, saldo resultant
e 789.00","fecha":"2026-10-23T05:34:30.637510Z"}]
PS C:\Users\dieg7\mexi-banco-s09> Remove-Item Env:FALLAR_ABONO_A_CLABE
PS C:\Users\dieg7\mexi-banco-s09> docker compose up -d transferencia
[+] up 4/4
 ✔Container mexi-banco-v06-rabbitmq-1      Healthy                                                                    8.7s
 ✔Container mexi-banco-v06-discovery-1     Healthy                                                                    8.7s
 ✔Container mexi-banco-v06-db-1            Healthy                                                                    8.7s
 ✔Container mexi-banco-v06-transferencia-1 Started                                                                    9.4s
PS C:\Users\dieg7\mexi-banco-s09>
```

Con la falla que se provocó en el abono a Beto, se intentó una transferencia de 150.

TAREA 5

```
PS C:\Users\dieg7\mexi-banco-s09> $env:PAUSA_ANTES_DEL_ACK_MS = "15000"
PS C:\Users\dieg7\mexi-banco-s09> docker compose up --build -d notificacion
[+] Building 10.6s (32/32) FINISHED
 => [internal] load local bake definitions                                                                           0.1s
 => => reading from stdin 1.06kB                                                                                     0.1s
 => [notificacion internal] load build definition from Dockerfile                                                    0.3s
 => => transferring dockerfile: 451B                                                                                 0.0s
 => [discovery internal] load build definition from Dockerfile                                                       0.2s
 => => transferring dockerfile: 490B                                                                                 0.0s
 => [discovery internal] load metadata for docker.io/library/eclipse-temurin:21-jdk-alpine                           2.5s
 => [notificacion internal] load metadata for docker.io/library/eclipse-temurin:21-jre-alpine                        2.4s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                               0.0s
 => [discovery internal] load .dockerignore                                                                          0.2s
 => => transferring context: 48B                                                                                     0.0s
 => [notificacion internal] load .dockerignore                                                                       0.3s
 => => transferring context: 48B                                                                                     0.0s
 => [notificacion internal] load build context                                                                       0.1s
 => => transferring context: 2.09kB                                                                                  0.0s
 => [notificacion builder 1/8] FROM docker.io/library/eclipse-temurin:21-jdk-alpine@sha256:fd52084c35bbac78890d95d57707d31ffc9  0.2s
 => => resolve docker.io/library/eclipse-temurin:21-jdk-alpine@sha256:fd52084c35bbac78890d95d57707d31ffc9d7a3324199e89c06be2a9  0.1s
 => [discovery stage-1 1/3] FROM docker.io/library/eclipse-temurin:21-jre-alpine@sha256:b48e02e0ea88d7bfb305335df94923bdef3949  0.1s
 => => resolve docker.io/library/eclipse-temurin:21-jre-alpine@sha256:b48e02e0ea88d7bfb305335df94923bdef394905f4147dccd52e873c  0.1s
 => [discovery internal] load build context                                                                          0.0s
 => => transferring context: 715B                                                                                    0.0s
 => CACHED [notificacion stage-1 2/3] WORKDIR /app                                                                   0.0s
 => CACHED [notificacion builder 2/8] WORKDIR /app                                                                   0.0s
 => CACHED [discovery builder 3/8] COPY pom.xml .                                                                    0.0s
 => CACHED [discovery builder 4/8] COPY mvnw .                                                                       0.0s
 => CACHED [discovery builder 5/8] COPY .mvn .mvn                                                                    0.0s
 => CACHED [discovery builder 6/8] RUN ./mvnw dependency:go-offline -B                                               0.0s
 => CACHED [discovery builder 7/8] COPY src ./src                                                                    0.0s
 => CACHED [discovery builder 8/8] RUN ./mvnw clean package -DskipTests                                              0.0s
 => CACHED [discovery stage-1 3/3] COPY --from=builder /app/target/*.jar app.jar                                     0.0s
 => CACHED [notificacion builder 3/8] COPY pom.xml .                                                                 0.0s
 => CACHED [notificacion builder 4/8] COPY mvnw .                                                                    0.0s
 => CACHED [notificacion builder 5/8] COPY .mvn .mvn                                                                 0.0s
 => CACHED [notificacion builder 6/8] RUN ./mvnw dependency:go-offline -B                                            0.0s
```

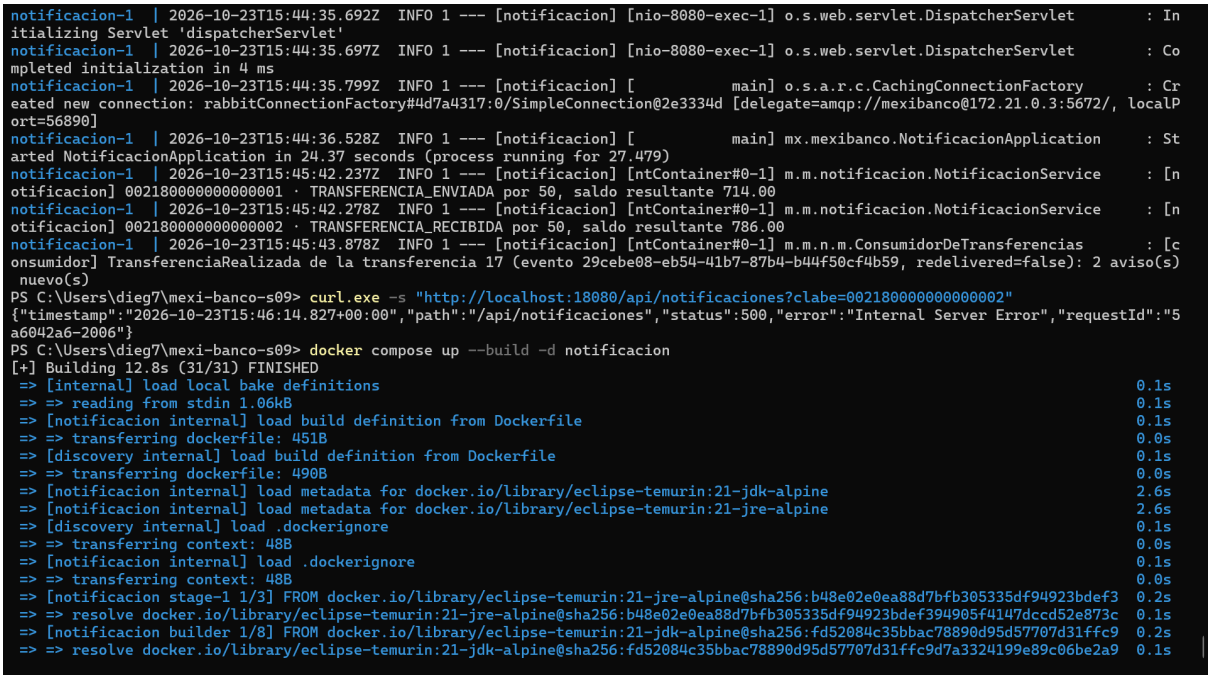
Se levanta notificacion con PAUSA_ANTES_DEL_ACK_MS=15000 y se hace una transferencia de 50 responde 201 COMPLETADA.

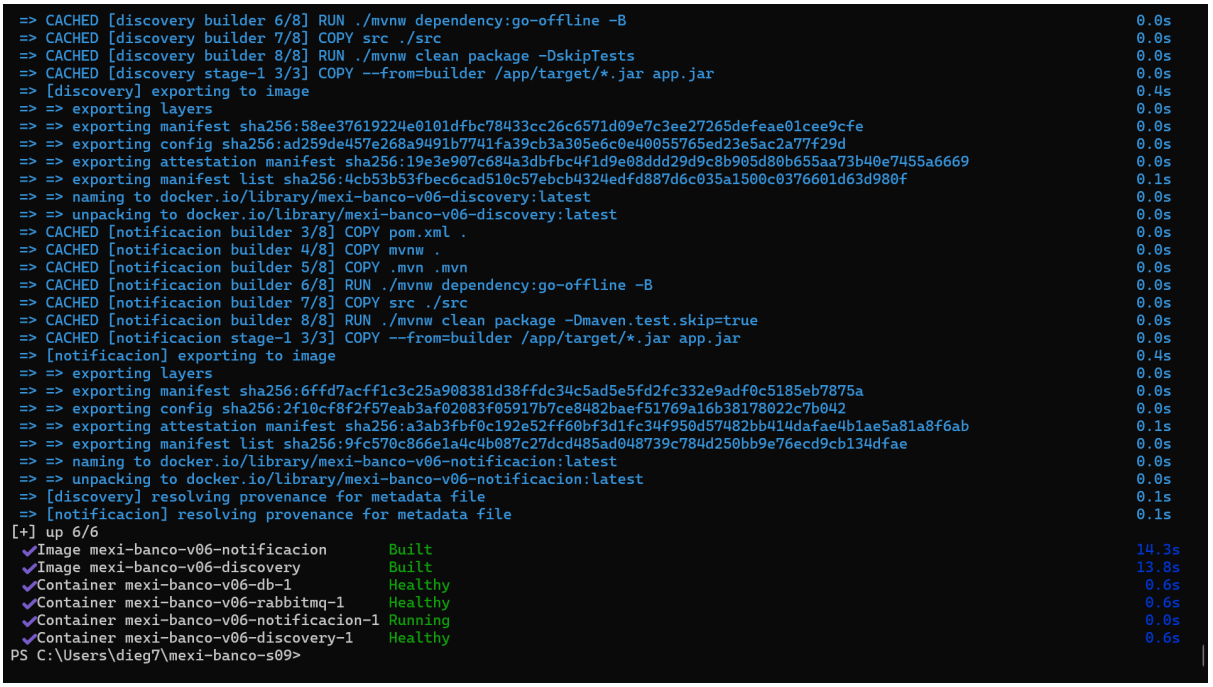
```
 => CACHED [notificacion builder 7/8] COPY src ./src                                                                0.0s
 => CACHED [notificacion builder 8/8] RUN ./mvnw clean package -Dmaven.test.skip=true                               0.0s
 => CACHED [notificacion stage-1 3/3] COPY --from=builder /app/target/*.jar app.jar                                 0.0s
 => [discovery] exporting to image                                                                                  0.2s
 => => exporting layers                                                                                             0.0s
 => => exporting manifest sha256:58ee37619224e0101dfbc78433cc26c6571d09e7c3ee27265defeae01cee9cfe                   0.0s
 => => exporting config sha256:ad259de457e268a9491b7741fa39cb3a305e6c0e40055765ed23e5ac2a77f29d                     0.0s
 => => exporting attestation manifest sha256:9f8ade6decac146c33c30e48c5df1699672cf27a235fef39f8f38681d14ee91a       0.1s
 => => exporting manifest list sha256:35e71495866e84e167cefe9a570a4b29656ed5d466cdae3a62aa5f5a29fce395              0.0s
 => => naming to docker.io/library/mexi-banco-v06-discovery:latest                                                  0.0s
 => => unpacking to docker.io/library/mexi-banco-v06-discovery:latest                                               0.0s
 => [notificacion] exporting to image                                                                               0.2s
 => => exporting layers                                                                                             0.0s
 => => exporting manifest sha256:6ffd7acff1c3c25a908381d38ffdc34c5ad5e5fd2fc332e9adf0c5185eb7875a                   0.0s
 => => exporting config sha256:2f10cf8f2f57eab3af02083f05917b7ce8482baef51769a16b38178022c7b042                     0.0s
 => => exporting attestation manifest sha256:e194bb8c9a49c54eebdaed1eee9e661044f67246385dc7afeee90093842676cd       0.0s
 => => exporting manifest list sha256:0929d2685d40e9c9081dd999d4c9ab8208d8b92c9ad609c6931799ddc453fe07              0.0s
 => => naming to docker.io/library/mexi-banco-v06-notificacion:latest                                               0.0s
 => => unpacking to docker.io/library/mexi-banco-v06-notificacion:latest                                            0.0s
 => [discovery] resolving provenance for metadata file                                                              0.0s
 => [notificacion] resolving provenance for metadata file                                                           0.0s
[+] up 6/6
 ✔Image mexi-banco-v06-notificacion       Built                                                                    11.4s
 ✔Image mexi-banco-v06-discovery          Built                                                                    11.4s
 ✔Container mexi-banco-v06-notificacion-1 Running                                                                   0.0s
 ✔Container mexi-banco-v06-rabbitmq-1     Healthy                                                                   0.6s
 ✔Container mexi-banco-v06-db-1           Healthy                                                                   0.6s
 ✔Container mexi-banco-v06-discovery-1    Healthy                                                                   0.6s
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -i -X POST http://localhost:18080/api/transferencias -H "Content-Type: application/json" -
d "@t50.json"
HTTP/1.1 201 Created
transfer-encoding: chunked
Content-Type: application/json
Date: Fri, 23 Oct 2026 15:45:38 GMT

{"id":17,"claveOrigen":"0021800000000000001","eventoId":"29cebe08-eb54-41b7-87b4-b44f50cf4b59","claveDestino":"0021800000000000002","mo
nto":50,"fecha":"2026-10-23T15:45:37.002333132Z","estado":"COMPLETADA","motivo":null,"saldoOrigenTrasCargo":714.00,"intentosDeCompens
```

Se detiene notificación con docker compose kill, simulando que muere a media operación, se quita la variable de la pausa y se vuelve a levantar.

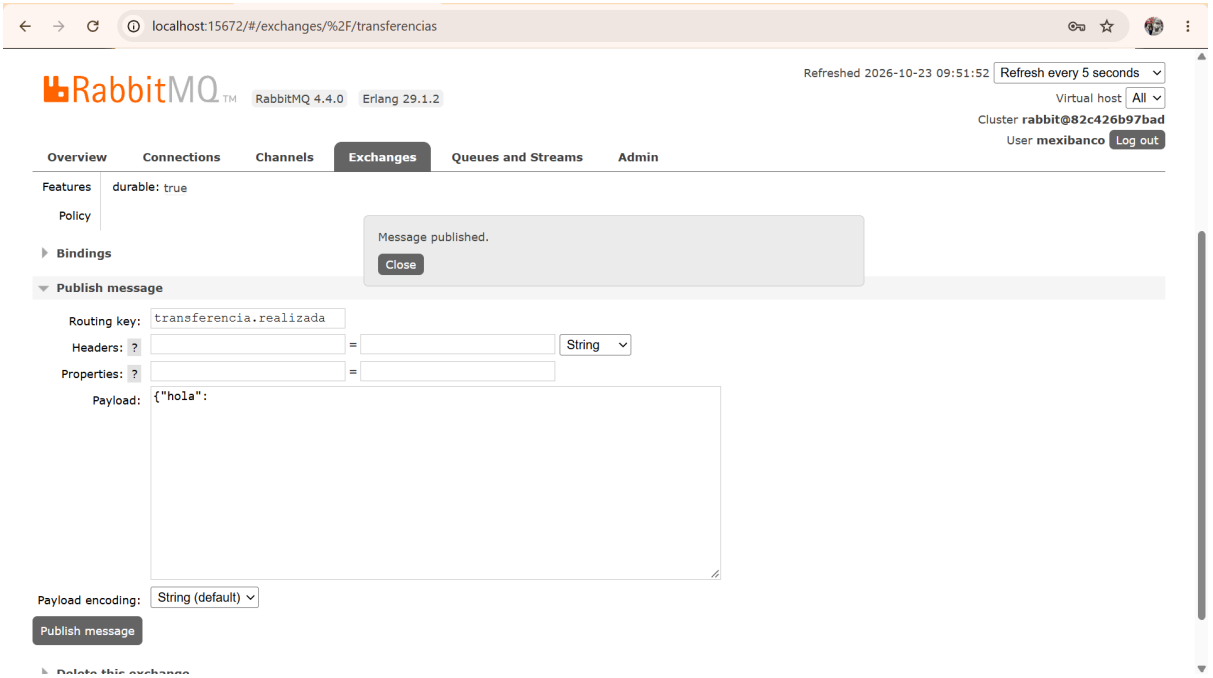
```
acion":0,"actualizada":"2026-10-23T15:45:39.202187348Z"}
PS C:\Users\dieg7\mexi-banco-s09> docker compose kill notificacion
[+] kill 1/1
 ✔Container mexi-banco-v06-notificacion-1 Killed                                                                    1.8s
PS C:\Users\dieg7\mexi-banco-s09> Remove-Item Env:PAUSA_ANTES_DEL_ACK_MS
PS C:\Users\dieg7\mexi-banco-s09> docker compose up -d notificacion
[+] up 4/4
 ✔Container mexi-banco-v06-rabbitmq-1     Healthy                                                                   0.6s
 ✔Container mexi-banco-v06-discovery-1    Healthy                                                                   0.6s
 ✔Container mexi-banco-v06-db-1           Healthy                                                                   0.6s
 ✔Container mexi-banco-v06-notificacion-1 Started                                                                   0.9s
PS C:\Users\dieg7\mexi-banco-s09> docker compose logs --tail 20 notificacion
notificacion-1  | 2026-10-23T15:44:32.091Z  INFO 1 --- [notificacion] [           main] com.netflix.discovery.DiscoveryClient    : Ge
tting all instance registry info from the eureka server
notificacion-1  | 2026-10-23T15:44:35.145Z  INFO 1 --- [notificacion] [           main] com.netflix.discovery.DiscoveryClient    : Th
e response status is 200
notificacion-1  | 2026-10-23T15:44:35.172Z  INFO 1 --- [notificacion] [           main] com.netflix.discovery.DiscoveryClient    : St
arting heartbeat executor: renew interval is: 5
notificacion-1  | 2026-10-23T15:44:35.190Z  INFO 1 --- [notificacion] [           main] c.n.discovery.InstanceInfoReplicator     : In
stanceInfoReplicator onDemand update allowed rate per min is 4
notificacion-1  | 2026-10-23T15:44:35.220Z  INFO 1 --- [notificacion] [           main] com.netflix.discovery.DiscoveryClient    : Di
scovery Client initialized at timestamp 1792770275213 with initial instances count: 4
notificacion-1  | 2026-10-23T15:44:35.306Z  INFO 1 --- [notificacion] [           main] o.s.c.n.e.s.EurekaServiceRegistry        : Re
gistering application NOTIFICACION with eureka with status UP
notificacion-1  | 2026-10-23T15:44:35.309Z  INFO 1 --- [notificacion] [           main] com.netflix.discovery.DiscoveryClient    : Sa
w local status change event StatusChangeEvent [timestamp=1792770275308, current=UP, previous=STARTING]
notificacion-1  | 2026-10-23T15:44:35.315Z  INFO 1 --- [notificacion] [foReplicator-%d] com.netflix.discovery.DiscoveryClient    : Di
scoveryClient_NOTIFICACION/6eedb99c3e08:notificacion:8080: registering service...
notificacion-1  | 2026-10-23T15:44:35.386Z  INFO 1 --- [notificacion] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : To
mcat started on port 8080 (http) with context path '/'
notificacion-1  | 2026-10-23T15:44:35.392Z  INFO 1 --- [notificacion] [           main] .s.c.n.e.s.EurekaAutoServiceRegistration : Up
dating port to 8080
notificacion-1  | 2026-10-23T15:44:35.417Z  INFO 1 --- [notificacion] [           main] o.s.a.r.c.CachingConnectionFactory       : At
tempting to connect to: [rabbitmq:5672]
notificacion-1  | 2026-10-23T15:44:35.562Z  INFO 1 --- [notificacion] [foReplicator-%d] com.netflix.discovery.DiscoveryClient    : Di
scoveryClient_NOTIFICACION/6eedb99c3e08:notificacion:8080 - registration status: 204
notificacion-1  | 2026-10-23T15:44:35.691Z  INFO 1 --- [notificacion] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/]       : In
```



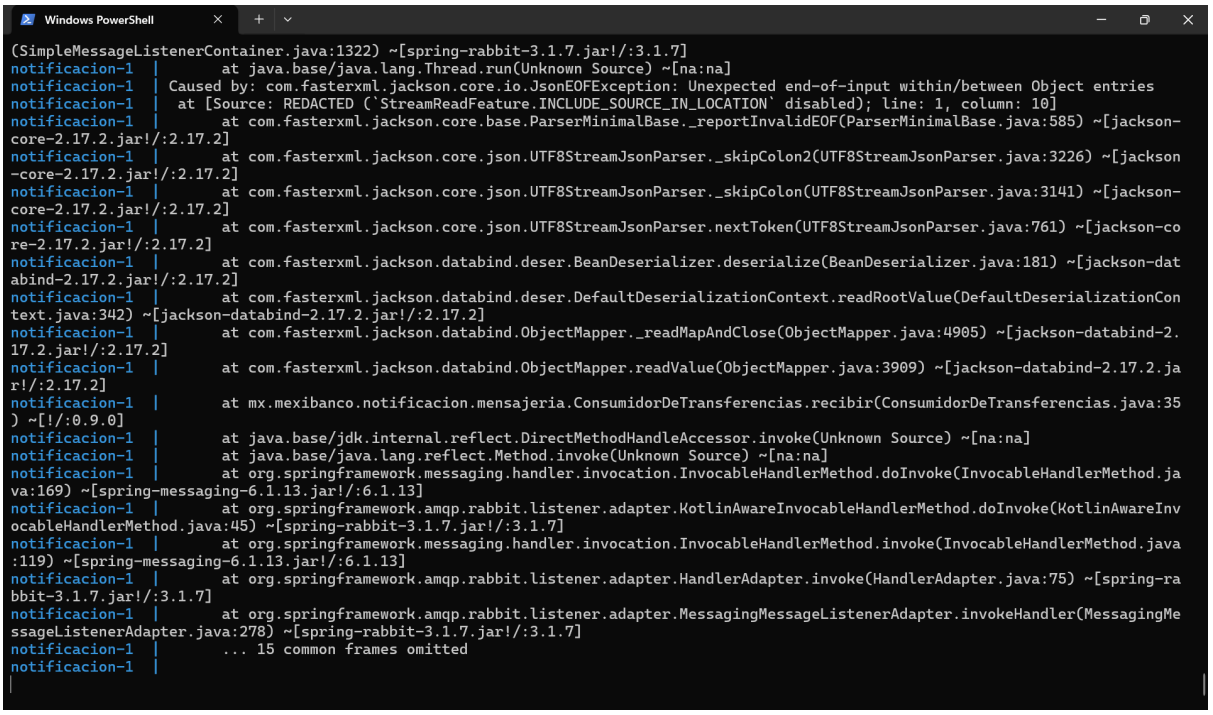


Al reiniciar, el consumidor procesa el evento de la transferencia 17 y guarda 2 avisos. La consulta de la bandeja respondió 500, esto sucede por haber consultado antes de que el servicio terminara de registrarse.

TAREA 6



Desde el panel se publica en el exchange transferencias, con routing key transferencia.realizada, el cuerpo {"hola":, que es un JSON incompleto y no un evento válido.

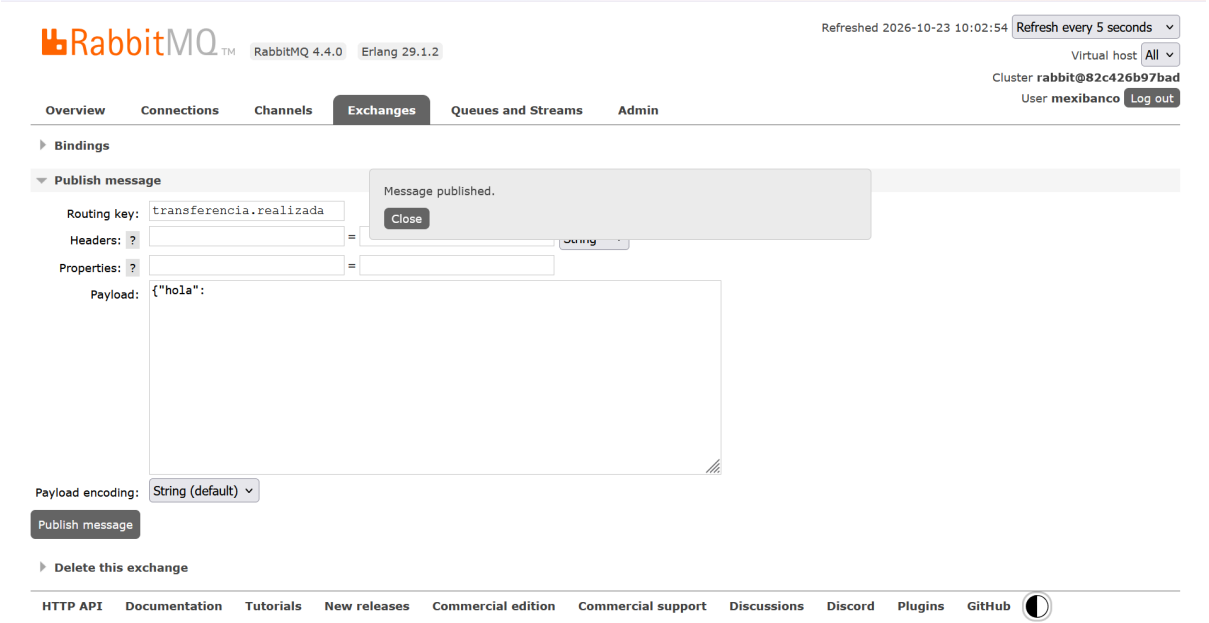


El consumidor falla con JsonEOFException: Unexpected end-of-input: no puede leer el mensaje porque el JSON está cortado. El mensaje se queda trabado.

```
PS C:\Users\dieg7\mexi-banco-s09> docker compose exec rabbitmq rabbitmqctl list_queues name messages_ready messages_unacknowledged
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages_ready  messages_unacknowledged
notificacion.transferencias     0       1
PS C:\Users\dieg7\mexi-banco-s09> docker compose exec rabbitmq rabbitmqctl delete_queue notificacion.transferencias
Deleting queue 'notificacion.transferencias' on vhost '/' ...
Queue was successfully deleted with 0 ready messages
PS C:\Users\dieg7\mexi-banco-s09> docker compose up --build -d notificacion
[+] Building 10.0s (32/32) FINISHED
 => [internal] load local bake definitions                                                                                0.1s
 => => reading from stdin 1.06kB                                                                                          0.1s
 => [discovery internal] load build definition from Dockerfile                                                            0.1s
 => => transferring dockerfile: 490B                                                                                      0.0s
 => [notificacion internal] load build definition from Dockerfile                                                         0.1s
 => => transferring dockerfile: 451B                                                                                      0.0s
 => [discovery internal] load metadata for docker.io/library/eclipse-temurin:21-jre-alpine                                4.3s
 => [discovery internal] load metadata for docker.io/library/eclipse-temurin:21-jdk-alpine                                4.3s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                    0.0s
 => [notificacion internal] load .dockerignore                                                                            0.0s
 => => transferring context: 48B                                                                                          0.0s
 => [discovery internal] load .dockerignore                                                                               0.0s
 => => transferring context: 48B                                                                                          0.0s
 => [notificacion stage-1 1/3] FROM docker.io/library/eclipse-temurin:21-jre-alpine@sha256:b48e02e0ea88d7bfb305335df94923bdef3  0.1s
 => => resolve docker.io/library/eclipse-temurin:21-jre-alpine@sha256:b48e02e0ea88d7bfb305335df94923bdef394905f4147dccd52e873c  0.1s
 => [discovery builder 1/8] FROM docker.io/library/eclipse-temurin:21-jdk-alpine@sha256:fd52084c35bbac78890d95d57707d31ffc9d7a  0.1s
 => => resolve docker.io/library/eclipse-temurin:21-jdk-alpine@sha256:fd52084c35bbac78890d95d57707d31ffc9d7a3324199e89c06be2a9  0.1s
 => [notificacion internal] load build context                                                                            0.0s
 => => transferring context: 2.09kB                                                                                       0.0s
 => [discovery internal] load build context                                                                               0.0s
 => => transferring context: 715B                                                                                         0.0s
 => CACHED [notificacion stage-1 2/3] WORKDIR /app                                                                        0.0s
 => CACHED [notificacion builder 2/8] WORKDIR /app                                                                        0.0s
 => CACHED [discovery builder 3/8] COPY pom.xml .                                                                         0.0s
 => CACHED [discovery builder 4/8] COPY mvnw .                                                                            0.0s
 => CACHED [discovery builder 5/8] COPY .mvn .mvn                                                                         0.0s
```

```
 => CACHED [discovery builder 6/8] RUN ./mvnw dependency:go-offline -B                                                    0.0s
 => CACHED [discovery builder 7/8] COPY src ./src                                                                         0.0s
 => CACHED [discovery builder 8/8] RUN ./mvnw clean package -DskipTests                                                   0.0s
 => CACHED [discovery stage-1 3/3] COPY --from=builder /app/target/*.jar app.jar                                          0.0s
 => CACHED [notificacion builder 3/8] COPY pom.xml .                                                                      0.0s
 => CACHED [notificacion builder 4/8] COPY mvnw .                                                                         0.0s
 => CACHED [notificacion builder 5/8] COPY .mvn .mvn                                                                      0.0s
 => CACHED [notificacion builder 6/8] RUN ./mvnw dependency:go-offline -B                                                 0.0s
 => CACHED [notificacion builder 7/8] COPY src ./src                                                                      0.0s
 => CACHED [notificacion builder 8/8] RUN ./mvnw clean package -Dmaven.test.skip=true                                     0.0s
 => CACHED [notificacion stage-1 3/3] COPY --from=builder /app/target/*.jar app.jar                                       0.0s
 => [notificacion] exporting to image                                                                                     0.3s
 => => exporting layers                                                                                                   0.0s
 => => exporting manifest sha256:6ffd7acff1c3c25a908381d38ffdc34c5ad5e5fd2fc332e9adf0c5185eb7875a                         0.0s
 => => exporting config sha256:2f10cf8f2f57eab3af02083f05917b7ce8482baef51769a16b38178022c7b042                           0.0s
 => => exporting attestation manifest sha256:c14b7f1c4d913f01dd63aff12d438cfc1b2cd91e21593b5c7f09c9dd3bdde81d             0.1s
 => => exporting manifest list sha256:9aeeb3ead8af2924d61e24f3491289b57d78f924a3f87e5f3a0ccf54faf53098                    0.1s
 => => naming to docker.io/library/mexi-banco-v06-notificacion:latest                                                     0.0s
 => => unpacking to docker.io/library/mexi-banco-v06-notificacion:latest                                                  0.0s
 => [discovery] exporting to image                                                                                        0.3s
 => => exporting layers                                                                                                   0.0s
 => => exporting manifest sha256:58ee37619224e0101dfbc78433cc26c6571d09e7c3ee27265defeae01cee9cfe                         0.0s
 => => exporting config sha256:ad259de457e268a9491b7741fa39cb3a305e6c0e40055765ed23e5ac2a77f29d                           0.0s
 => => exporting attestation manifest sha256:435128ec62e5ec500517f784028f471c3e233c9b84f9f00bf510524b61e100e9             0.1s
 => => exporting manifest list sha256:b4f475e6a2dec1e5a3661f3821d46587c7a6351fc0b8bccd505d8ddf2df06884                    0.1s
 => => naming to docker.io/library/mexi-banco-v06-discovery:latest                                                        0.0s
 => => unpacking to docker.io/library/mexi-banco-v06-discovery:latest                                                     0.0s
 => [notificacion] resolving provenance for metadata file                                                                 0.1s
 => [discovery] resolving provenance for metadata file                                                                    0.1s
[+] up 6/6
 ✔Image mexi-banco-v06-notificacion       Built                                                                         10.7s
 ✔Image mexi-banco-v06-discovery          Built                                                                         10.6s
 ✔Container mexi-banco-v06-discovery-1    Healthy                                                                        0.6s
 ✔Container mexi-banco-v06-notificacion-1 Running                                                                        0.0s
 ✔Container mexi-banco-v06-db-1           Healthy                                                                        0.9s
 ✔Container mexi-banco-v06-rabbitmq-1     Healthy                                                                        0.6s
PS C:\Users\dieg7\mexi-banco-s09>
```

rabbitmqctl muestra 0 mensajes Ready y 1 Unacked el mensaje malo quedó trabado en la cola. Después se borra la cola y se reconstruye notificación, porque RabbitMQ no permite cambiar los argumentos de una cola que ya existe.



Se publica otra vez {"hola": para comprobar el comportamiento después del arreglo.

```
PS C:\Users\dieg7\mexi-banco-s09> docker compose exec rabbitmq rabbitmqctl list_queues name messages_ready messages_unacknowledged
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages_ready  messages_unacknowledged
notificacion.transferencias     0       1
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -i -X POST http://localhost:18080/api/transferencias -H "Content-Type: application/json" -
d "@t50.json"
HTTP/1.1 201 Created
transfer-encoding: chunked
Content-Type: application/json
Date: Fri, 23 Oct 2026 16:03:42 GMT

{"id":18,"claveOrigen":"00218000000000001","eventoId":"47b8502e-45f8-47e4-bf12-e938181a138d","claveDestino":"00218000000000002","mo
nto":50,"fecha":"2026-10-23T16:03:40.8427934302","estado":"COMPLETADA","motivo":null,"saldoOrigenTrasCargo":664.00,"intentosDeCompens
acion":0,"actualizada":"2026-10-23T16:03:42.660301919Z"}
PS C:\Users\dieg7\mexi-banco-s09> curl.exe -s "http://localhost:18080/api/notificaciones?clabe=00218000000000002"
[{"id":22,"clabeCuenta":"00218000000000002","mensaje":"TRANSFERENCIA_RECIBIDA por 50, saldo resultante 836.00","fecha":"2026-10-23T1
6:03:44.299569Z"},{"id":20,"clabeCuenta":"00218000000000002","mensaje":"TRANSFERENCIA_RECIBIDA por 50, saldo resultante 786.00","fec
ha":"2026-10-23T15:45:42.278412Z"},{"id":18,"clabeCuenta":"00218000000000002","mensaje":"TRANSFERENCIA_RECIBIDA por 13, saldo result
ante 736.00","fecha":"2026-10-23T05:34:30.759477Z"},{"id":16,"clabeCuenta":"00218000000000002","mensaje":"TRANSFERENCIA_RECIBIDA por
 12, saldo resultante 723.00","fecha":"2026-10-23T05:34:30.752276Z"},{"id":14,"clabeCuenta":"00218000000000002","mensaje":"TRANSFERE
NCIA_RECIBIDA por 11, saldo resultante 711.00","fecha":"2026-10-23T05:34:30.637664Z"}]
PS C:\Users\dieg7\mexi-banco-s09>
```

Una transferencia normal sigue respondiendo 201 y generando avisos en la bandeja de Beto.